

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

___ show ip route ospf ___

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ S0/0/0 ___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___ 129 ___

Does R2 show up as an OSPF neighbor on R1? ___ Yes ___

Does R2 show up as an OSPF neighbor on R3? ___ No ___

What does this information tell you?

___ All traffic from R3 to 192.168.2.0/24 is going through R1. ___

___ R2 is still passive on S0/0/1, so R3 cannot form an OSPF adjacency with R2. ___

___ Therefore R3 reaches 192.168.2.0/24 through R1, with cost $129 = 64 + 64 + 1$. ___

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

___ R2(config)# router ospf 1 ___

___ R2(config-router)# no passive-interface s0/0/1 ___

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ S0/0/1 ___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

___ $65 = 64 + 1$, that is one serial T1 link cost (64) plus the R2 G0/0 LAN cost (1). ___

Is R2 listed as an OSPF neighbor to R3? ___ Yes ___

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

___ Modern links are often faster than 100 Mb/s, so the default value is too low. ___

___ Raising the reference bandwidth makes OSPF cost calculation more accurate on high-speed links. ___

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

___ Cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 = 781 + 1 = 782. ___

___ Cost to 192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 = 781 + 64 = 845. ___

___ So one route uses one serial link and one LAN, while the other uses two serial links. ___

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___ 1,562. After setting all serial interfaces to bandwidth 128, each serial link cost becomes 781. ___

___ The route to 192.168.23.0/24 crosses two serial links, so 781 + 781 = 1,562. ___

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___ OSPF always chooses the route with the lowest accumulated cost. ___

___ Via R2, the cost is 781 + 781 + 1 = 1,563. ___

___ The direct path to R3 is 1,565 + 1 = 1,566, so the route through R2 is preferred. ___

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?
___ The router ID identifies the router in OSPF and affects DR/BDR election. ___
___ If it is learned from an active interface, it may change when that interface goes down. ___
___ That can cause instability or confusion in OSPF operation. ___
___ So it is better to set it manually with a loopback address or the router-id command. ___
2. Why is the DR/BDR election process not a concern in this lab?
___ DR/BDR election matters only on multi-access networks such as Ethernet. ___
___ In this lab, the inter-router links are serial point-to-point links. ___
___ Point-to-point OSPF links do not perform DR/BDR election. ___
3. Why would you want to set an OSPF interface to passive?
___ To stop unnecessary OSPF hello and update traffic where no OSPF neighbor should form. ___
___ This saves bandwidth and reduces unnecessary protocol overhead. ___
___ The router still advertises that connected network through its other active OSPF interfaces. ___
4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?
___ OSPF is hierarchical because it divides an autonomous system into areas. ___
___ The backbone area 0 and area border routers make the network more scalable. ___
5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?
___ network 10.0.0.0 0.255.255.255 area 0 ___
___ This wildcard mask matches all interfaces from 10.0.0.0 to 10.255.255.255. ___
6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?
___ The EIGRP route will be used. ___
___ Route selection uses administrative distance: EIGRP 90, OSPF 110, RIP 120. ___